

Extra-oral Radiographic Techniques

	Lateral Cephal	SMV	Waters	PA Cephal	Reverse Towne	Schickel Lateral Body	Flamant
Patient placement	Film parallel to midsagittal plane	Carthormetal line parallel to film	Carthormetal line at 37° with film	Carthormetal line at 10° with film	Carthormetal line at -30° with film	Film in contact with cheek at molar area	Film in contact with cheek at ramus area
Central beam	Beam perpendicular to film	Beam perpendicular to film	Beam perpendicular to film	Beam perpendicular to film	Beam perpendicular to film	Beam aims at the molar-premolar area	Beam aims at the ramus area
Diagram of patient placement							
Illustration of patient placement							
Skull view							
Resultant image							

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Indications

- Fractures of the maxillofacial skeleton
- Fractures of the skull
- Investigation of the Antra
- Diseases affecting the Skull base and Vault
- TMJ disorders
- Cyst and tumors of maxillofacial region
- Osteomyelitis of jaw bones
- Growth and development
- Mixed dentition analysis



Mandibular Oblique Views

[Body, Ramus]

tilted the head obliquely to the affected side

- Indications:
 - Assessment of the presence and/or position of unerupted teeth
 - Detection of fractures of the mandible
 - Evaluation of lesions or conditions of the jaws including cysts, tumours, giant cell lesions, and osteodystrophies
 - As an alternative when intraoral views [gagging or if the patient is unable to open mouth, Unconscious]
 - As specific views of the salivary glands or TMJ

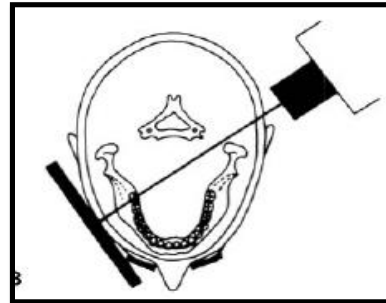
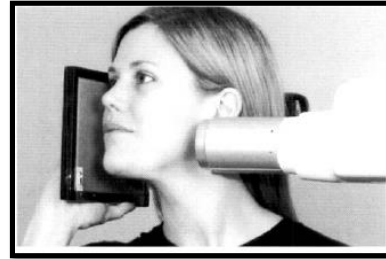


Technique and Positioning: Oblique Mandibular

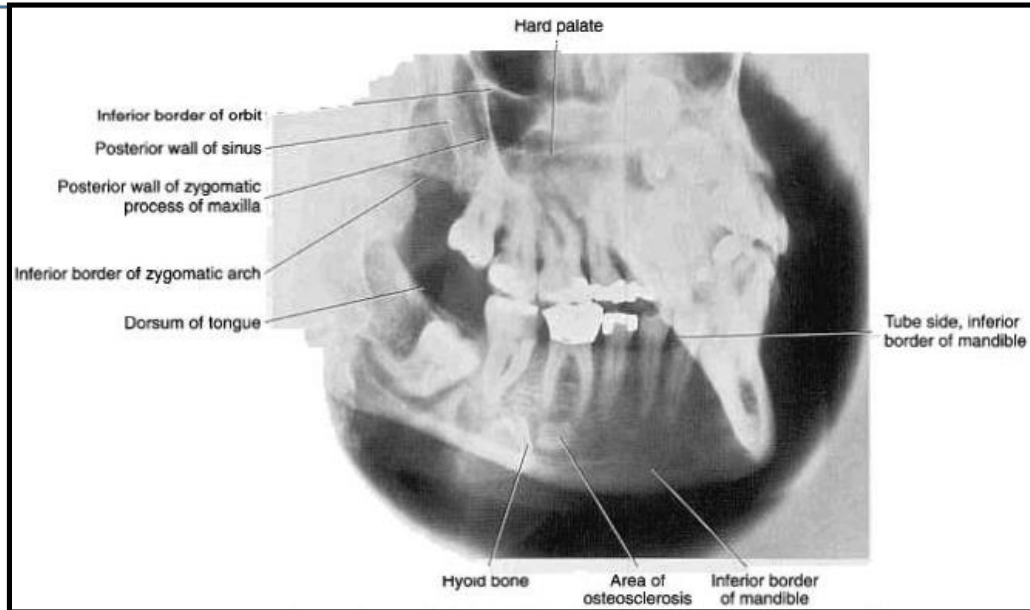
Body view

and mark angle of the mandible 2cm below

- The cassette is held by the patient against the side of the face overlying the area of interest
- *Rotate the head to the side of interest*
[avoiding its superimposition of Ramus and increase the space between the neck and shoulder to position the X-ray tube]
- Central beam aimed at Molar- premolar region 2cm below the Opposite Side Angle of Mandible

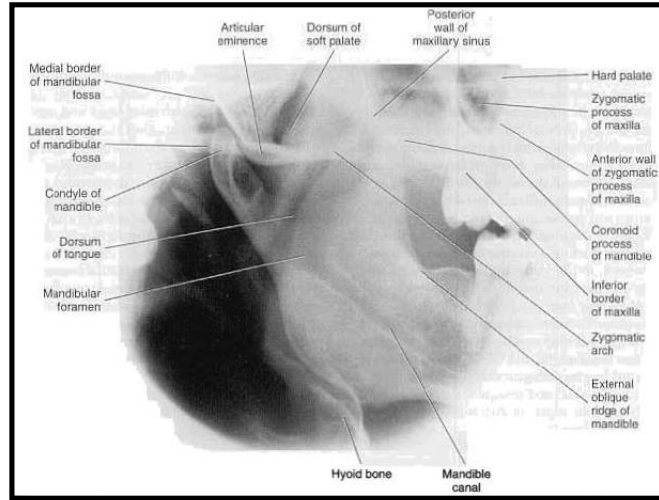


- Resultant Image



Technique and Positioning: Oblique Mandibular Ramus View

- The central beam is directed toward the center of the imaged ramus, from 2 cm below the Inferior Border of the opposite side of the Mandible



indication
TMJ problem

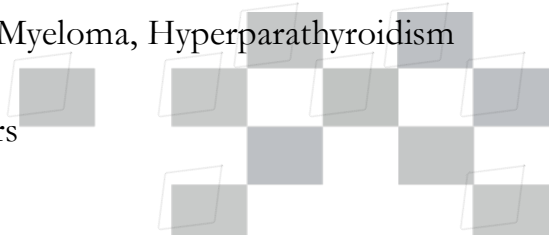


Lateral Skull

right, left side of skull

Indications:

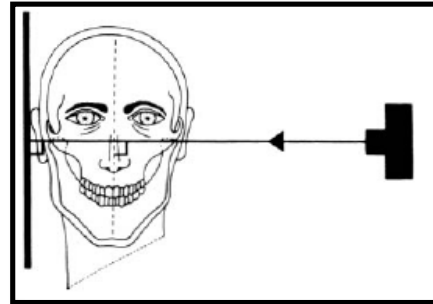
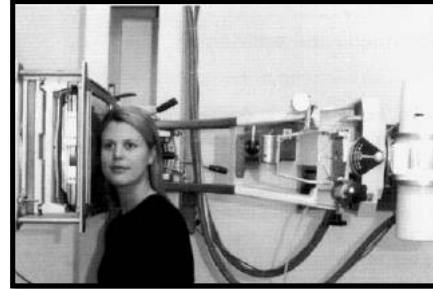
- Fractures of the cranium and the cranial base
- Middle 1/3 facial fractures: show possible downward and backward displacement of the Maxillae
- Investigation of the Frontal, Sphenoidal and Maxillary sinuses
- Conditions affecting the skull vault: Paget's disease, Multiple Myeloma, Hyperparathyroidism
- Conditions affecting the Sella turcica: Pituitary gland Tumours



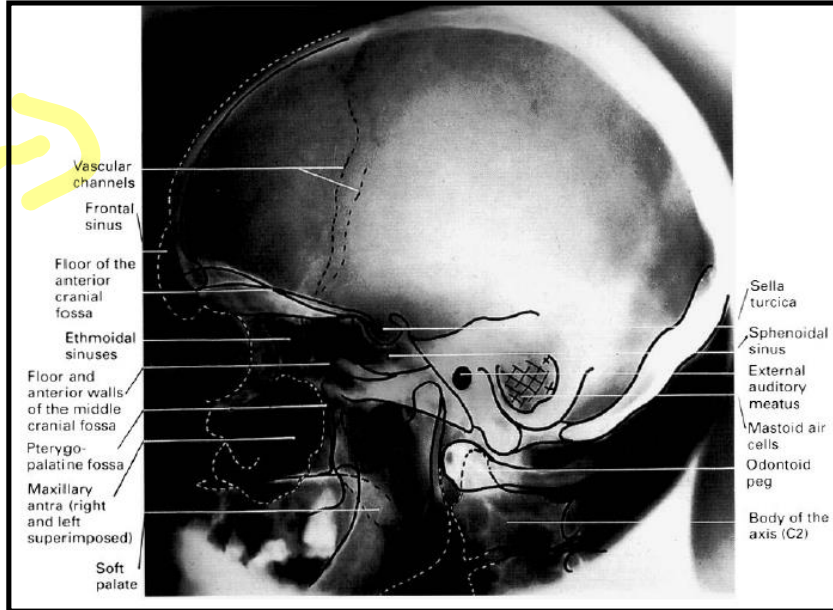
Technique and Positioning

- Patient Position: Image Receptor is positioned Parallel to the patient's Mid-sagittal Plane
- Position of the Central X-Ray Beam: Perpendicular to the Mid-sagittal plane of the patient and the plane of the image receptor, and centered over the External Auditory Meatus

orthodontic treatment



■ Resultant Image:



Sub-mento Vertex [SMV]

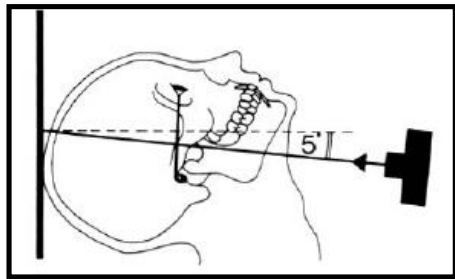
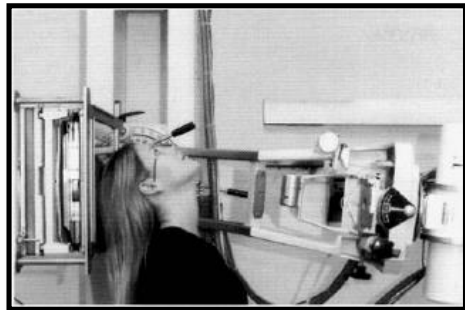
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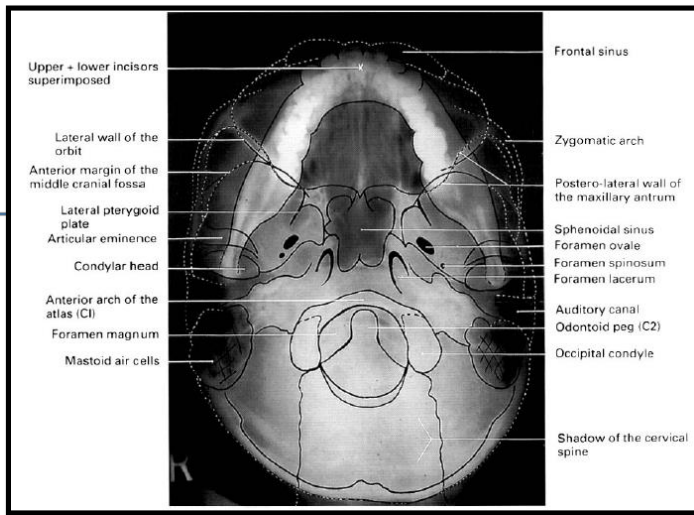
- Destructive/expansive lesions affecting the palate, pterygoid region or base of skull
- Investigation of the sphenoidal sinus
- Assessment of the thickness [medio-lateral] of the posterior part of the mandible before osteotomy
- Fracture of the zygomatic arches: to show these thin bones the SMV is taken with reduced exposure factors



Technique and Positioning

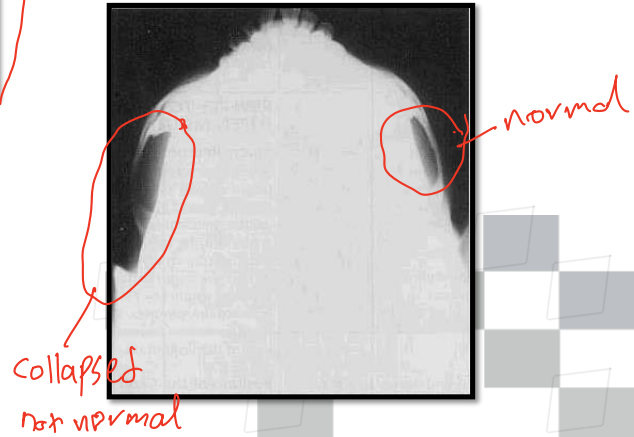
- **Patient positioning:** Head is tipped backwards as far as is possible, so the vertex of the skull touches the film
- **X-ray Tubehead:** Aimed upwards from below the chin, with the Central Ray at 5° to the horizontal line
- **Centered on an Imaginary line joining the lower First Molars**





- Sub-mento vertex showing
Only the Zygomatic Arches

Fracture easy



Water's Projection

Use for:

- 1- for all paranasal sinuses
- 2- EN (ear, nose)

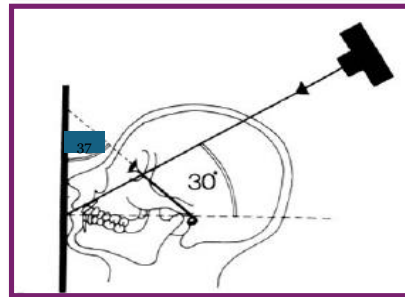
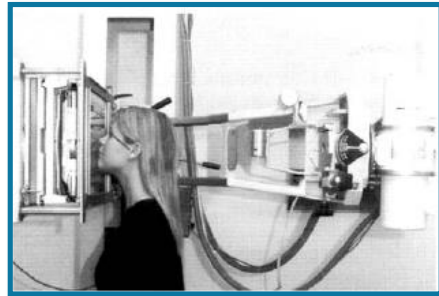
- Indications:
- Detecting the following middle third facial fractures: LeFort fractures, Coronoid process fractures



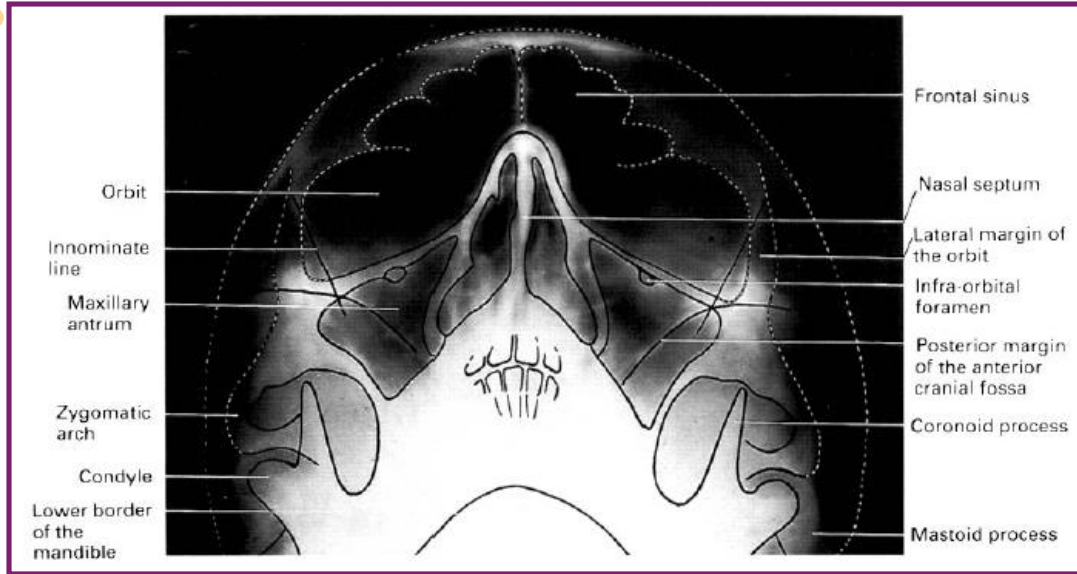
Technique and Positioning

also called occipito menta 37°

- The image receptor is placed perpendicular to the midsagittal plane of patient and
- The patient's head is tilted upward: The Cantho-meatal line forms a 37° angle with the image receptor, in “Nose-Chin” position
- The central beam is perpendicular to the image receptor and centered in the area of the Maxillary Sinuses



- Resultant image



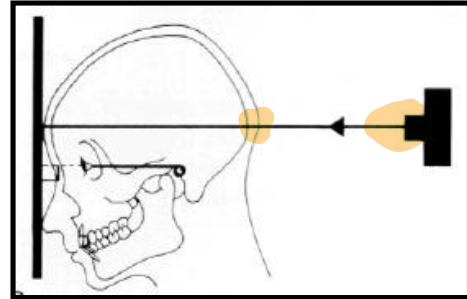
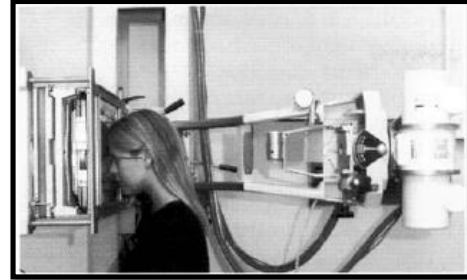
Postero-anterior skull projection

- Indications:
 - Fractures of the skull vault
 - Investigation of the frontal sinuses
 - Conditions affecting the cranium, particularly:
 - Paget's disease
 - multiple myeloma
 - hyperparathyroidism
 - Intracranial calcification



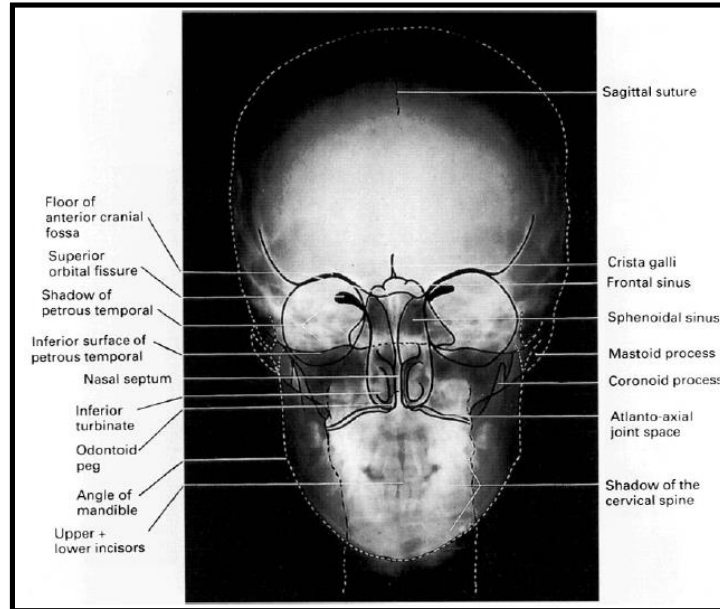
Positioning and Technique

- The patient: Positioned facing the film with the head tipped forwards, the so-called “Forehead-Nose” Position
- The X-ray tubehead is positioned with the Central Ray Horizontal [0°] Centred through the “Occiput”



- Resultant image

from front



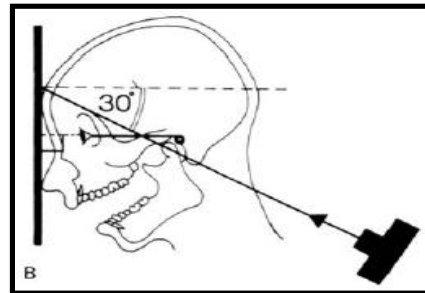
Reverse- towne projection [open mouth]

- Indications:
- High fractures of the condylar necks
- Intracapsular fractures of the TMJ
- Investigation of the articular surfaces of the Condylar Heads in TMJ
[Condylar hypoplasia or hyperplasia]

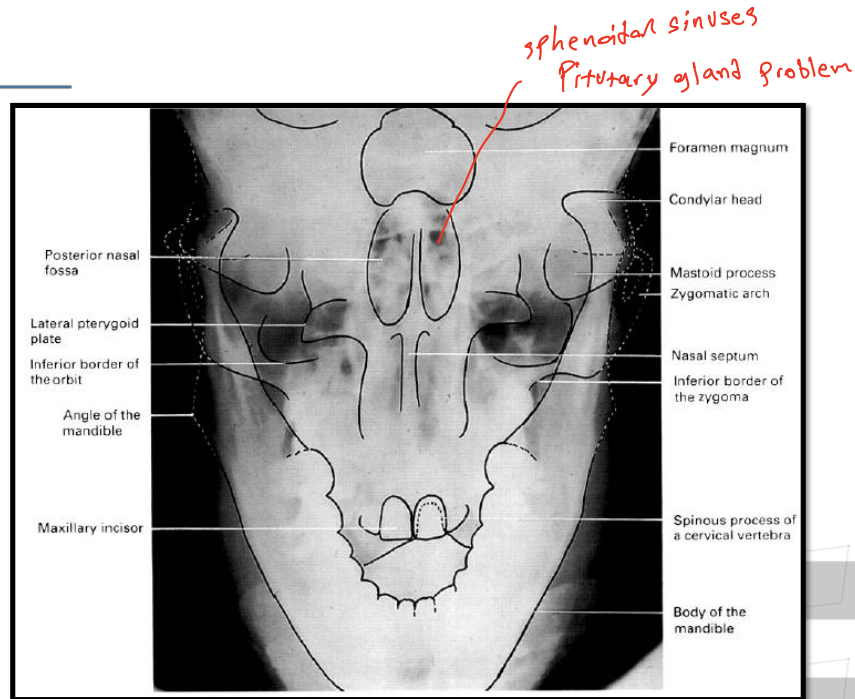


Technique and Positioning

- The patient: Positioned facing the film with the head tipped forwards, "Forehead-Nose" Position with "Open Mouth" [takes the Condylar Heads out of the Glenoid Fossae so they can be seen]
- The X-ray tubehead is aimed Upwards from below the Occiput, with the Central Ray at 30° to the horizontal, Centred through the Condyles



- Resultant image



Thank You

